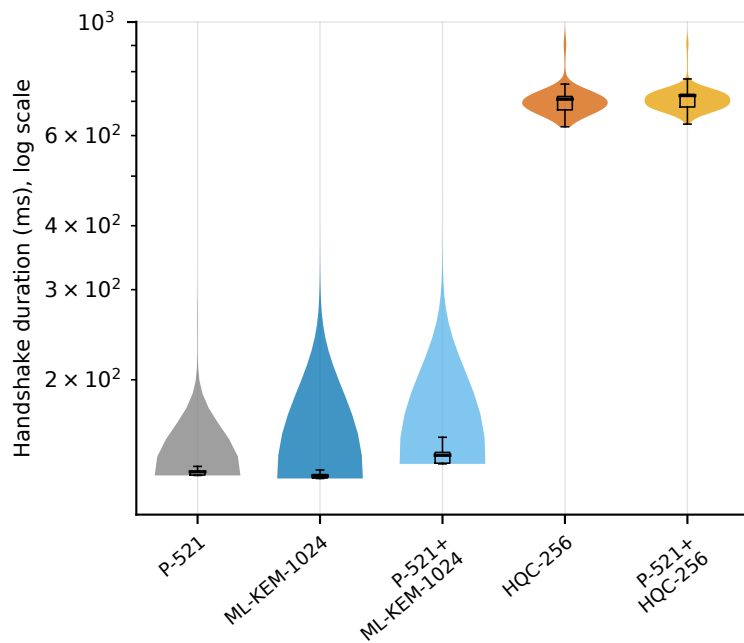
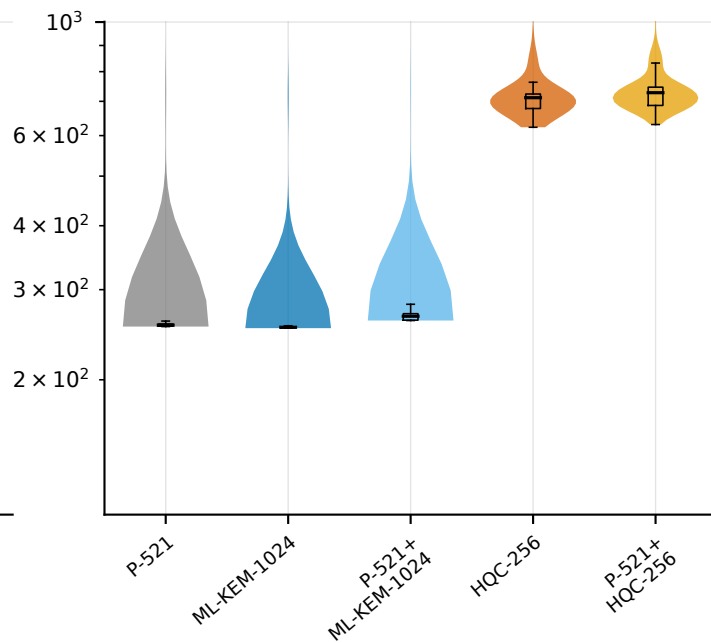


(a)

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**

Classical KEX ML-KEM (pure) ML-KEM (hybrid) HQC (pure) HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)
P-521	141.0	65.8	1.27k	291.9	104.5	3.40k
ML-KEM-1024	156.2	119.5	1.53k	281.4	81.2	2.50k
P-521+ML-KEM-1024	171.6	117.1	1.85k	304.8	97.7	4.00k
HQC-256	715.1	13.4	1.36k	738.7	19.7	1.78k
P-521+HQC-256	715.0	10.1	1.35k	745.6	14.2	1.96k